Datasets: Simple Guide for the Team

Project: Skin Cancer Detection Using Deep Learning

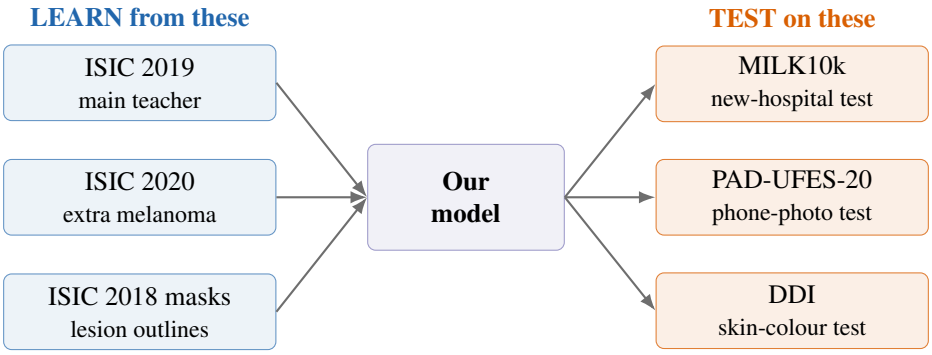
1. What is a dataset?

A dataset is a big folder of skin photos where doctors have already written the correct answer (the *label*) for each photo, for example "melanoma" or "harmless mole". Our model learns by looking at thousands of these labelled photos.

2. Words you need to know

Word	Simple meaning
Lesion	A spot, mole, or patch on the skin
Malignant	Cancer (dangerous)
Benign	Not cancer (harmless)
Melanoma (MEL)	The most dangerous skin cancer
BCC, SCC	The two most common skin cancers
Actinic keratosis (AK)	A rough patch that can turn into cancer
Dermoscopic image	A close-up photo taken with a special skin magnifier
Clinical / smartphone image	A normal photo taken with a camera or phone
Mask	A black-and-white image that marks exactly where the lesion is
Metadata	Extra patient details: age, sex, body part

3. The datasets we will use (the plan in one picture)



Why two groups? If we test the model on the same kind of photos it learned from, it is like giving a student the exact exam questions they practised. To prove the model really works, we test it on photos from *other* hospitals, *other* cameras, and *other* skin colours.

4. Datasets for LEARNING (training)

ISIC 2019 — our main dataset

The biggest collection of labelled skin-lesion photos with many classes.

Size	25,331 photos
What is inside	8 classes: melanoma, BCC, SCC, actinic keratosis (these 4 are cancer or pre-cancer), and 4 harmless types. Also age, sex, body part.
How we use it	Main training data. It already contains HAM10000 and BCN20000, so we do **not** download those again.
Link	https://challenge.isic-archive.com/data/

ISIC 2020 (SIIM-ISIC Melanoma)

Photos from 2,056 real patients, mostly harmless moles.

Size	33,126 photos (only 584 are melanoma)
What is inside	Melanoma vs. not melanoma, plus age, sex, body part, patient ID.
How we use it	Extra melanoma examples and practice with rare cancer, like in real life.
Link	https://www.kaggle.com/c/siim-isic-melanoma-classification

ISIC 2018 Task 1 — lesion masks

Photos where experts have drawn the exact outline of each lesion.

Size	2,594 photos with masks
What is inside	Photo + mask pairs.
How we use it	Train the segmentation model that finds and cuts out the lesion.
Link	https://challenge.isic-archive.com/data/

5. Datasets for TESTING

MILK10k (ISIC 2025) — new-hospital test

The newest ISIC dataset. Each lesion has a normal photo *and* a close-up photo.

Size	5,240 lesions (10,480 photos)
What is inside	11 classes including melanoma, BCC, SCC. Almost all confirmed by lab test (biopsy). Includes skin tone.
How we use it	Final test: does our model work on photos it has never seen? (Remove any photo that is also in ISIC 2019 first.)
Link	https://doi.org/10.34970/648456

PAD-UFES-20 — phone-photo test

Photos taken with ordinary smartphones in Brazil.

Size	2,298 photos
What is inside	6 classes: BCC, SCC, melanoma, and 3 non-cancer types. Includes skin type.
How we use it	Test: does a model trained on close-up photos still work on phone photos?
Link	https://doi.org/10.17632/zr7vgbcyr2.1

DDI — skin-colour fairness test

Photos from Stanford with an equal mix of light, medium, and dark skin.

Size	656 photos
What is inside	Cancer vs. not cancer, all confirmed by biopsy, with skin-tone group.
How we use it	Test: is our model equally good for dark and light skin? (Needs a free sign-up.)
Link	https://ddi-dataset.github.io/

PH2 — small outline test

A small, high-quality set from Portugal.

Size	200 photos
What is inside	Moles and melanomas, with expert outlines.
How we use it	Test the segmentation model on a different hospital.
Link	https://www.fc.up.pt/addi/ph2%20database.html

6. Other datasets (good to know, not required)

Name	Photos	Note	Link
HAM10000	10,015	Most famous dataset; already inside ISIC 2019	doi.org/10.7910/DVN/DBW86T
BCN20000	18,946	Hard cases from Barcelona; already inside ISIC 2019	doi.org/10.6084/m9.figshare.24140028
SLICE-3D (ISIC 2024)	401,059	Huge, but only 0.1% cancer; for future work	kaggle.com/competitions/isic-2024-challenge
Fitzpatrick 17k	16,577	Many skin diseases with skin-colour labels	github.com/mattgroh/fitzpatrick17k
Derm7pt	1,011 cases	Doctor's 7-point checklist labels	derm.cs.sfu.ca
ISIC Archive	–	Search and download any ISIC image	isic-archive.com

7. Five simple rules for using the data

1. **Remove copies.** Some photos appear in more than one dataset. Find and delete duplicates before anything else.
2. **Keep the same lesion together.** If one mole has 3 photos, all 3 must go into training *or* all 3 into testing, never split.
3. **Never train on the test datasets.** MILK10k, PAD-UFES-20, and DDI are only for the final exam.
4. **Use the same class names.** Different datasets use different names for the same thing (AK, AKIEC, ACK). Make one list and convert all labels to it.
5. **Cancer photos are rare.** Do not judge the model by accuracy alone; check how many cancers it catches.

Summary: Learn from ISIC 2019 + ISIC 2020 (+ ISIC 2018 masks). Test on MILK10k, PAD-UFES-20, DDI, and PH2. All are free for student research. The full detailed version is in `Datasets_Report_Detailed.pdf`.